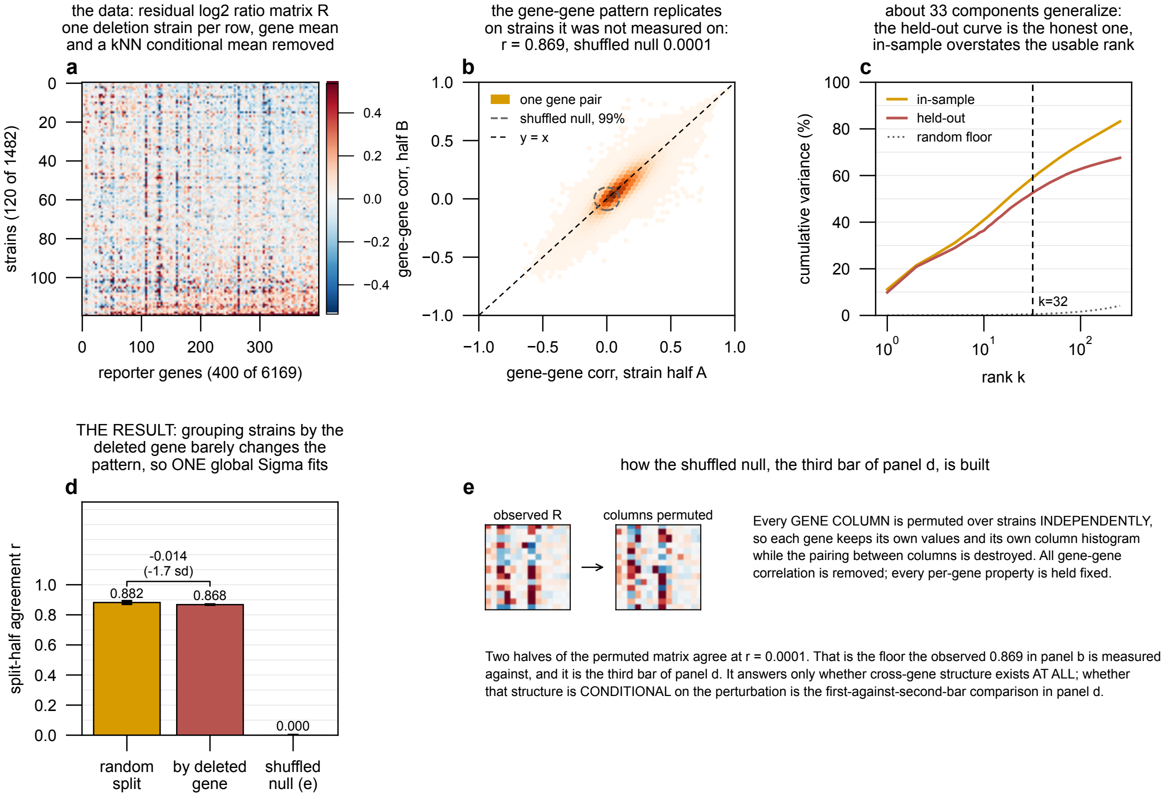
the data: residual log2 ratio matrix R
one deletion strain per row, gene mean
and a kNN conditional mean removed
a
strains (120 of 1482)
reporter genes (400 of 6169)
0
20
40
60
80
100
0
100
200
300
0.4
0.2
0.0
−0.2
−0.4
the gene-gene pattern replicates
on strains it was not measured on:
r = 0.869, shuffled null 0.0001
b
one gene pair
shuffled null, 99%
y = x
gene-gene corr, half B
gene-gene corr, strain half A
1.0
0.5
0.0
−0.5
−1.0
about 33 components generalize:
the held-out curve is the honest one,
in-sample overstates the usable rank
c
in-sample
held-out
random floor
k=32
cumulative variance (%)
rank k
100
80
60
40
20
0
10^0
10^1
10^2
THE RESULT: grouping strains by the
deleted gene barely changes the
pattern, so ONE global Sigma fits
d
split-half agreement r
-0.014
(-1.7 sd)
0.882
0.868
0.000
1.0
0.8
0.6
0.4
0.2
0.0
random split
by deleted gene
shuffled null (e)
how the shuffled null, the third bar of panel d, is built
e
observed R
columns permuted
Every GENE COLUMN is permuted over strains INDEPENDENTLY, so each gene keeps its own values and its own column histogram while the pairing between columns is destroyed. All gene-gene correlation is removed; every per-gene property is held fixed.
Two halves of the permuted matrix agree at r = 0.0001. That is the floor the observed 0.869 in panel b is measured against, and it is the third bar of panel d. It answers only whether cross-gene structure exists AT ALL; whether that structure is CONDITIONAL on the perturbation is the first-against-second-bar comparison in panel d.